

Solutions: Assignment 1 — Python Setup and Using ChatGPT for Debugging

A. Setup Guidance

Option 1 — Anaconda: Install Anaconda, open Anaconda Navigator, launch Jupyter Lab.

Option 2 — Python + pip: Install Python 3.10+, then run:

```
pip install jupyterlab  
jupyter lab
```

Verification cells:

```
print("Hello, Python World!")  
5 + 3
```

B. Debugging Walkthroughs

Broken Script A → Errors: missing colon; missing argument.

Suggested ChatGPT prompt:

```
"Here is my code and error. Please explain the traceback and give  
the minimal fix with a quick test."
```

Fixed code:

```
def add_numbers(a, b):  
    return a + b  
  
print(add_numbers(5, 3))  
assert add_numbers(2,2) == 4
```

Broken Script B → Error: missing colon; style improvement.

Fixed code:

```
items = ["apple", "banana", "cherry"]  
for item in items:  
    print(item)  
assert len(items) == 3
```

Broken Script C → Error: type issue; clarify percent vs decimal.

Fixed code (percent assumed):

```
rate = "5"  
months = 12  
rate_pct = float(rate) / 100.0  
monthly = 10000 * (rate_pct/12)  
print(round(monthly, 2))  
assert round(10000 * ((5/100)/12), 2) == round(monthly, 2)
```

C. Evaluation Checklist — Sample Answers

- All corrected snippets run without errors.
- Fixes are minimal (typos, colons, type conversions).
- No hallucinated libraries used.
- Simple asserts validate the results.